

Internship Test: Python, Data Science & Machine Learning

Duration: 2 hours

Instructions:

- You are required to complete both sections of the test.
- Make sure to submit your code for each question.
- Your answers will be evaluated based on correctness, logic, and efficiency.
- **Do not use ChatGPT or any other AI tools** to complete the tasks. Your submission must reflect your own work.
- Submit your test on this email: **coderroots002@gmail.com**

Section 1: Python for Data Science

Task 1: Data Manipulation and Cleaning

Given a CSV file named `employee_data.csv` with the following columns: EmployeeID, Name, Department, Salary, JoiningDate, Status. Download the dataset from here:

https://github.com/gurminder07/test_datasets/blob/main/employee_data.csv

1. **Data Cleaning:** Write Python code to:
 - Remove any duplicate entries.
 - Handle missing values (you can choose to fill them with a default value or drop the rows).
 - Convert the **JoiningDate** column to a proper datetime format.
 - Filter out employees who are no longer working (where **Status** is 'Resigned').
 - Create a new column `YearsOfService = CurrentYear(Take this as 2025) - JoiningDate`.
2. **Analysis:** After cleaning the data, perform the following:
 - Find the average salary of employees in each department.
 - List all employees who joined after the year 2020.
3. **Expected Output:** Submit a cleaned DataFrame and the results of your analysis.

Task 2: Data Visualization

Using a public dataset of your choice (for example, Kaggle datasets or any other source), perform the following:

1. Load the dataset into a Pandas DataFrame.
2. Create at least four meaningful visualizations using **Matplotlib** or **seaborn** to understand the data better.
3. Include one correlation heatmap of all numerical variables in the dataset.

Expected Output: Submit the code, the dataset, and the visualizations.

Section 2: Machine Learning

Task 3: Predictive Modeling (Classification)

You are provided with the **Heart Disease dataset**. Your task is to build a classification model to predict the presence of heart disease in a patient based on various health metrics.

1. **Dataset Information:**

- Download the Heart Disease dataset from the UCI repository.
Direct link: https://github.com/gurminder07/test_datasets/blob/main/heart_data.csv
- This dataset contains 14 attributes such as age, sex, chest pain type, resting blood pressure, cholesterol level, and more. The target variable is target(0 indicates no heart disease, 1- indicates heart disease).

2. **Data Preprocessing:**

- Replace any missing or undefined values.
- Convert categorical variables into numerical (using techniques such as one-hot encoding or label encoding).
- Normalize or scale features if necessary.

3. **Task:** Build a machine learning classification model to predict heart disease (use binary classification where 0 = no heart disease, 1 = presence of heart disease).

- Split the data into training and testing sets.
- Train the model using one of the following algorithms:
 - Logistic Regression
 - Decision Tree
 - Random Forest
- Evaluate your model using accuracy, precision, recall, and F1 score.

4. **Expected Output:** Submit the code, evaluation metrics, and a brief explanation of why you chose this specific model.

Section 3: Natural Language Processing (NLP)

Task 4: Sentiment Analysis (NLP)

Dataset: Use the [Sentiment Analysis Dataset](#)

1. **Data Preprocessing:**

- Clean the text data (remove stopwords, punctuation, and special characters).
- Convert the text into numerical format using TF-IDF or Word Embedding (e.g., Word2Vec).

2. **Tasks:**

- Build a sentiment analysis model to classify text as Positive or Negative.
- Split the dataset into training and testing sets.
- Train a classification model using any of the following:
 - Simple RNN
 - LSTM RNN
- Evaluate the model using **accuracy, precision, recall, and F1 score**.

Expected Output: Submit the cleaned dataset, code, and evaluation results.

Task 4:GitHub Upload(Bonus Task)

1. Create a public GitHub repository for this project.
 2. Upload all your code, data, and documentation (README.md with instructions) to this repository.
 3. Make sure the repository is well-organized and includes:
 - A clear project description.
 - Installation instructions and how to run the code.
 - Any dependencies or libraries used.
 4. Share the link to the GitHub repository as part of your submission.
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Submission Requirements

- Python scripts or Jupyter Notebooks (.py or .ipynb files).
- Include comments in your code to explain your logic.
- Submit all files (data, code, visualizations) as a zip folder or share a link to a repository.